



**CHAITANYA BHARATHI
INSTITUTE OF TECHNOLOGY (A)**

Kokapet(Village), Gandipat, Hyderabad, Telangana-500075. www.cbilt.ac.in



Course Code: 16CSE14

Course Title: CYBER SECURITY

Instructors Name:

Name	Email
M Venkata Krishna Reddy	krishnareddy_cse@cbilt.ac.in

Objective:

The primary goals of the module/unit are to introduce:

- Cyber Crime - Legal Perspective – World Wide and in Indian Scenario
- About Cyber Crimes.
- Cyber Law and its importance.
- Indian IT Act
- Cyber Crime, Cyber Law and Punishments.
- Digital Signatures and Influence.

Outcome:

On Successful completion of the course, students will be able to understand about Cyber Crime, Legal Perspectives in Indian Scenario, able to gain knowledge on Cyber Laws, their impact, Indian Act, able to know the punishments and about Digital Signatures.

Scope:

Cyber Security: The Legal Perspectives: Cyber Crime and the Legal Landscape around the World, Need of Cyber laws: the Indian Context, The Indian IT Act, Challenges to Indian Law and Cyber Crime Scenario in India, Digital Signatures and the Indian IT Act, Cyber Crime and Punishment, Cyber Law, Technology and Students: The Indian Scenario.

Text and Reference:

(a) Text Book:

1. **TB1.** Sunit Belpre and Nina Godbole, “Cyber Security: Understanding Cyber Crimes, Computer Forensics And Legal Perspectives”, Wiley India Pvt.Ltd, 2011.
2. **TB2.** Kevin Mandia, Chris Prosis, “Incident Response and computer forensics”, Tata McGraw Hill, 2006.

b) Reference Books:

1. **RB1.** Alfred Basta, Nadine Basta, Mary Brown, Ravinder Kumar, “Cyber Security and Cyber Laws”, Paperback – 2018.

Lecture Notes :

The Legal Perspectives: Cyber Crime and the Legal Landscape around the World :

Nowadays, the term cyber crime is well known and needs no introduction. Crime is a great hurdle in the development of a country. It adversely affects the members of the society and lowers down the economic growth of the country. Computer technology provides a boost to the human life and makes it easier and comfortable. It adds accuracy, speed and efficiency to the life of human being. But a computer is exploited by the criminals and its illegal use leads to cyber crime. To combat cyber crime, India enacted the Information Technology Act, 2008 which was drastically amended in the year 2008 providing more powerful and stringent law. Cyber crime is a crime done with the misuse of information technology for unauthorized or illegal access, electronic fraud; like deletion, alteration, interception, concealment of data, forgery etc.. Cyber crime is an international crime as it has been affected by the global revolution in information and communication technologies (ICTs). It has affected the global community. It would be unlawful act where the The internet has become an integral part of everyone's life. It has also given new dimensions to our economic and social life. But at the same time we cannot be oblivious of the negative side of use of computers and internet. It is very unfortunate that computer crime is rampant and is increasing exponentially as the side effect of the excessive use of computers and internet. The internet security problem is immensely growing and cyber crimes are continuously increasing even though we are using many countermeasures.

Need of Cyber laws :

Today, it cannot be overemphasized that billions of users are using internet. The internet is used almost everywhere like in home, shop, office, railway station, college etc. by the users. The internet and our economies have also become interwoven. The internet generates both wealth and employment. Unfortunately, the internet is misused by hackers and organised criminals. The growth of cyber crime is increasing proportionately to the internet explosion. Cyber crime is expanding parallel with the growing number of internet users. The internet is open to the public and the internet users are at risk and targeted for mental harassment, financial gain through malware and social evil purposes. Therefore, for detection and prevention of such cyber threat, the industries are developing a range of products for use in the home and the business, for example, intrusion detection systems, firewalls, antivirus software etc.. Despite all the preventive

steps, we are not able to get rid of cyber crime. The internet security problem is immensely growing and cyber crime continues to thrive even though we are using many countermeasures. Due to these consequences there was need to adopt a strict law by the cyber space authority to regulate criminal activities relating to cyber and to provide better administration of justice to the victim of cyber crime. In the modern cyber technology world, it is very much necessary to regulate cyber crimes and most importantly cyber law should be made stricter in the case of cyber terrorism and hackers.

The Indian IT Act :

The Information Technology Act, 2000 also Known as an **IT Act** is an act proposed by the Indian Parliament reported on 17th October 2000. This Information Technology Act is based on the United Nations Model law on Electronic Commerce 1996 (UNCITRAL Model) which was suggested by the General Assembly of United Nations by a resolution dated on 30th January, 1997. It is the most important law in India dealing with Cybercrime and E-Commerce. The main objective of this act is to carry lawful and trustworthy electronic, digital and online transactions and alleviate or reduce cybercrimes. The IT Act has 13 chapters and 90 sections. The last four sections that starts from 'section 91 – section 94', deals with the revisions to the Indian Penal Code 1860.

The IT Act, 2000 has two schedules:

- **First Schedule :**

Deals with documents to which the Act shall not apply.

- **Second Schedule :**

Deals with electronic signature or electronic authentication method.

The offences and the punishments in IT Act 2000 :

The offences and the punishments that falls under the IT Act, 2000 are as follows :-

1. Tampering with the computer source documents.
2. Directions of Controller to a subscriber to extend facilities to decrypt information.
3. Publishing of information which is obscene in electronic form.
4. Penalty for breach of confidentiality and privacy.
5. Hacking for malicious purposes.
6. Penalty for publishing Digital Signature Certificate false in certain particulars.

7. Penalty for misrepresentation.
8. Confiscation.
9. Power to investigate offences.
10. Protected System.
11. Penalties for confiscation not to interfere with other punishments.
12. Act to apply for offence or contravention committed outside India.
13. Publication for fraud purposes.
14. Power of Controller to give directions.

Sections and Punishments under Information Technology Act, 2000 are as follows :

Cyber Law (IT Law) in India

Cyber Law also called IT Law is the law regarding Information-technology including computers and internet. It is related to legal informatics and supervises the digital circulation of information, software, information security and e-commerce.

IT law does not consist a separate area of law rather it encloses aspects of contract, intellectual property, privacy and data protection laws. Intellectual property is a key element of IT law. The area of software license is controversial and still evolving in Europe and elsewhere.

1. It covers all transaction over internet.
2. It keeps eyes on all activities over internet.
3. It touches every action and every reaction in cyberspace.

Area of Cyber Law:

Cyber laws contain different types of purposes. Some laws create rules for how individuals and companies may use computers and the internet while some laws protect people from becoming the victims of crime through unscrupulous activities on the internet. The major areas of cyber law include:

1. *Fraud:*

Consumers depend on cyber laws to protect them from online fraud. Laws are made to prevent identity theft, credit card theft and other financial crimes that happen online. A person who commits identity theft may face confederate or state criminal charges. They might also encounter a civil action brought by a victim. Cyber lawyers work to both defend and prosecute against allegations of fraud using the internet.

2. ***Copyright:***

The internet has made copyright violations easier. In early days of online communication, copyright violations was too easy. Both companies and individuals need lawyers to bring actions to impose copyright protections. Copyright violation is an area of cyber law that protects the rights of individuals and companies to profit from their own creative works.

3. ***Defamation:***

Several personnel use the internet to speak their mind. When people use the internet to say things that are not true, it can cross the line into defamation. Defamation laws are civil laws that save individuals from fake public statements that can harm a business or someone's personal reputation. When people use the internet to make statements that violate civil laws, that is called Defamation law.

4. ***Harassment and Stalking:***

Sometimes online statements can violate criminal laws that forbid harassment and stalking. When a person makes threatening statements again and again about someone else online, there is violation of both civil and criminal laws. Cyber lawyers both prosecute and defend people when stalking occurs using the internet and other forms of electronic communication.

5. ***Freedom of Speech:***

Freedom of speech is an important area of cyber law. Even though cyber laws forbid certain behaviors online, freedom of speech laws also allow people to speak their minds. Cyber lawyers must advise their clients on the limits of free speech including laws that prohibit obscenity. Cyber lawyers may also defend their clients when there is a debate about whether their actions consist of permissible free speech.

6. ***Trade Secrets:***

Companies doing businesses online often depend on cyber laws to protect their trade secrets. For example, Google and other online search engines spend lots of time developing the algorithms that produce search results. They also spend a great deal of time developing other features like maps, intelligent assistance and flight search services to name a few. Cyber laws help these companies to take legal action as necessary in order to protect their trade secrets.

7. ***Contracts and Employment Law:***

Every time you click a button that says you agree to the terms and conditions of using a

website, you have used cyber law. There are terms and conditions for every website that are somehow related to privacy concerns.

Advantages of Cyber Law:

- Organizations are now able to carry out e-commerce using the legal infrastructure provided by the Act.
- Digital signatures have been given legal validity and sanction in the Act.
- It has opened the doors for the entry of corporate companies for issuing Digital Signatures Certificates in the business of being Certifying Authorities.
- It allows Government to issue notification on the web thus heralding e-governance.
- It gives authority to the companies or organizations to file any form, application or any other document with any office, authority, body or agency owned or controlled by the suitable Government in e-form by means of such e-form as may be prescribed by the suitable Government.
- The IT Act also addresses the important issues of security, which are so critical to the success of electronic transactions.

CYBERSECURITY IN THE INDIAN LEGAL SCENARIO :

Status of Cyber Laws in India and its Importance :

Crime is defined to be a social and economic event. Cybercrime is not clearly defined under the Information Technology Act 2000, the amendment of the I.T. Act of 2008, or in the legislation in India. Under Indian jurisdiction, due to its lack of proper definition, cybercrime is said to be a combination of two words i.e. crime and computer. In proper terminology, any crime or offence committed using a computer or electronic device is categorised as cybercrime. The I.T. Act defines terms like a computer, computer network, data, information and all other necessary components that form part of cybercrime. In a cybercrime, a computer or its data itself is the target or the object of the offence in committing some other offence. The internet world is defined as cyberspace and the laws which prevail in this are known as Cyber laws. Cyber laws are described as a branch of laws that deal with the legal issues and conflicts in the internet world, and internet users are bound to follow this code of jurisdiction. Cyber laws in India can be considered as a part of the overall legal system that deals with the Internet, E-commerce, digital

contracts, electronic evidence, cyberspace, and their respective legal issues. Cyber laws cover a broad range of topics including freedom of expression, data protection, data security, digital transactions, electronic communication, access to and usage of the Internet, and online privacy.

The IT Act 2000 was implemented to amend some of the outdated laws, and it has helped in keeping the jurisdiction over cybercrimes. Cyber laws are important for people to perform online transactions without fear of misuse. The Act offers the much-needed legal framework so that information is not denied legal effect, validity or enforceability, solely on the ground that it is in the form of electronic records.

Need for an Exclusive Cyber security Framework

There have been several cyber attacks in India, especially during the COVID-19 lockdown. The cyber attacks have soared to about 86% during March and April. This led to the work on overhauling the 2013 cyber security policy. National **Cyber Security Policy** is a **policy** framework introduced in 2013 by the Department of Electronics and Information Technology. The main aim is to protect the public and private infrastructure from **cyber** attacks. It was important for the policy to be organized and strategized properly. Prime Minister Narendra Modi, in his speech on Independence Day 2020, announced that the Government is cognizant of the cyber attacks on India and its potential threats. Hence, he announced that there needs to be an immediate implementation of new cyber security policy.

The National Cyber Security Strategy (2020) aims at securing business data, which is a critical information infrastructure that can impact the security of the nation and the economy. New challenges include data protection/privacy, law enforcement in evolving cyberspace, access to data stored overseas, misuse of social media platforms, international cooperation on cybercrime and cyber terrorism, and so on. The National Cyber Security Strategy 2020 is needed because the existing structure needs to be revamped and revitalized.

Public and Private Synergy

Public-private partnership is important for formulating these policies and regulations and creating a cyber security education program for the private sector and common masses. Cyber security should be made compulsory for all the critical sectors of the Indian economy such as defense and national security. The financial institutions require an important legal framework and formalized precautionary measures in case of any relapse. Pawan Duggal, a cyber law expert said that cyber policy should have a holistic approach. There is scant clarity on the operating procedures in the public and private sectors. For a national approach to deal with cyber

challenges, a regional team of officers, from across the country, needs to be formed to deal with cyber security. The specialized cyber cells of every state would formulate better strategies comprising officers who have domain knowledge. These are some of the steps discussed by the Central Unit in brainstorming sessions for a new strategy.

Social Media Cybercrimes

Social media these days has become a key tool in displaying and spreading information to the public which is vital. Based on various sources, it was found that Pakistan and China have been spreading misinformation on social media. This was clearly visible during the ongoing tussle between India and China at the Line of Actual Control in Ladakh. A study proclaimed that social media platforms were used for spreading propaganda that caused public disorder. There is a need to surveil motivated use of social media by vested interests for disturbing law and order. The government has decided to take steps such as the introduction of artificial intelligence to remove objectionable content from social media platforms.

There was a state-sponsored hacking group supported by Pakistan which perpetrated a series of ransomware and phishing attacks aimed at stealing highly sensitive defense, security and diplomatic information under the guise of coronavirus health advisories. Scores of Indian government officials in India and its embassies around the world found themselves squarely in the crosshairs of the attacks masquerading as legitimate epidemic-related communication.

Covid-19 and Cybercrime

The spike of cybercrimes and attacks have increased during the pandemic. There has been a sharp attack on financial transaction information of citizens. Lt. Gen. Rajesh Pant, India's National Cyber Security Coordinator (NCSC), told the Economic Times that cybercriminals had launched thousands of "fraud portals" related to the coronavirus. These sites have lured thousands of Indians, eager to contribute to the fight against coronavirus, into making donations. Many of these phoney sites are quite sophisticated and virtually indistinguishable from their genuine counterparts. The PM cares fund initiative by the office of the Prime Minister for fighting Coronavirus had multiple fraudulent websites. A study reveals that there have been several malware and phishing schemes operating under the pretext of COVID prevention efforts. The so-called "coronavirus malware" is aimed at stealing bank account details, password and other sensitive information from users. India's historic lockdown has led to cybercrime beyond just opportunistic attacks against private individuals.

Need for Legal reforms

The Information and Technology Act, 2000 provides a framework for the security of electronic governance by giving identification to electronic records and digital signatures. It also defines penalties for cybercriminals. It has statutorily established a Cyber Appellate Tribunal to resolve the disputes related to Cyber Crimes and Online Frauds.

Additionally, some of the statutory provisions of Indian Penal Code, 1860 in the crime of Fraud, Criminal Intimidation, Cheating, Breach of Trust, Abetment of Suicide via Blackmailing, etc. may be charged on an accused considering the circumstances and discretionary powers of the Court. However, it must be noted that a person cannot be punished twice for the same offence as it will violate his Fundamental Right ensured under Article 20 (2) of the Indian Constitution i.e. Double Jeopardy.

Challenges to Indian Law and Cyber Crime Scenario in India :

Cyber security poses bigger threat than any other spectrum of technology. Cyber criminals have already started abusing technology controlled devices for propelling cyber-crimes such as frauds and thefts. With technology protocols, still being developed and evolving at a gradual pace, it is very difficult to avoid such cyber-attacks. IoT plays a dramatic role in shaping the future of technology in India. With IoT now becoming backbone of various ventures, firms, organization and even basic ways of living, it is worrying that India has no dedicated law for IoT and some kind of guidance can be referred from the Information Technology Act, 2000 (IT Act, 2000). The Digital India initiative is driving our country towards a digitized life where the existence will highly depend on elements like cloud computing, 5G in telecom, e-Commerce etc. it is imperative to keep a check on loose ends.

Few challenges that the technology space faces in cyber security are the following:

- **Digital Data Threat:** Growing online transactions generate bigger incentives for cybercriminals. Besides, establishments looking to mine data—for instance, customer information, results of product surveys, and generic market information—create treasured intellectual property that is in itself an attractive target.

- **Supply Chain Inter-connection:** The supply chains are increasingly interconnected. Companies are urging vendors and customers to join their networks. This makes a company's security wall thin.
- **Hacking:** This action is penetrating into someone's system in unauthorized fashion to steal or destroy data, which has grown hundred folds in the past few years. The availability of information online makes it easier for even non-technical people to perform hacking.
- **Phishing:** The easiest to execute and can produce the results with very little effort. It is the act of sending out Fake emails, text messages and create websites to look like they're from authentic companies.

These challenges can be under surveillance and methodical steps can be taken to avoid such malpractices. To solve data theft problem, online space must regulate the use of data and clearly indicate when information will be shared provided by the users. The user can then choose to opt out, leaving personal information restricted to the space for which it was deliberated. When software online contains bugs or viruses, it is fairly easy for cyber criminals to gain personal information. Large technology firms should collaborate and create solutions that to increase security for their customers. Security controls need to move outward, beginning at the application level where such frauds can be caught easily. When there are no unified monitoring methods, firms become vulnerable. However, when every network has monitoring that detects changes, data can be protected.

With cloud hovering our lives with all our data it is imperative to protect our cloud space. With growing technology, the growth of cybercrime is evident but measure taken early and effectively can avoid cyber mishaps both big and small.

Digital Signatures and the Indian IT Act :

Digital Signatures in India :

Electronic documents need to be signed to assure recipients of their authenticity, and digital signatures fulfill this need. Dominic K reports (Oct 2006 issue)

Digital signatures have been in use for quite a while to authenticate various e-commerce and m-commerce transactions. Today, the processes of creating and verifying a digital signature provide

a high level of assurance to the involved parties that the e-signature is genuinely the signer's, and that the electronic document (or the e-contract) is authentic.

A digital foundation of trust

Digital signatures are nothing but a cryptographic (encrypted) signature assurance scheme that lets both parties (sender and receiver) trust an electronic document and treat it as valid and tamper-proof as long as the said document stays in an electronic format.

To get technical, according to ISO/IEC 7498-2, a digital signature is defined as “data appended to, or a cryptographic transformation of a data unit, that allows the recipient of a data unit to prove the source and integrity of the data unit and protect against forgery.”

For individuals

A digital signature involves two components-the public key and the private key. The sender signs a document using his private key that ensures the document's safety in transit as the text is encrypted and only the sender has access to his private key. Therefore, by signing a document with it, he authenticates that it has originated with him and not been tampered with en route. The recipient of this document uses the sender's public key to authenticate the encrypted document and to decrypt it into a readable text format.

There are several ways to authenticate a person or the information on a computer. Some of them are password, checksum, CRC (cyclic redundancy check), private key encryption, public key encryption and digital certificate.

IT Act 2000

The Indian Information Technology Act 2000 ('Act') came into effect from October 17, 2000. The Act is by and large based on the United Nations Commission on International Trade Law (UNCITRAL) model law on electronic commerce.

The objective of the Act is to provide for legal recognition of electronic transactions and digital signatures. Section 5 of the Act gives legal recognition to digital signatures. Digital signatures have been legalized in India since 2000. However, since then, hardly any provisions of the Act have been implemented, except for the appointment of the Certifying Authority which took place in 2001.

A Few Provisions of IT Act 2000

Legal recognition of digital signatures (section 5). “Where any law provides that information or any other matter shall be authenticated by affixing the signature, or any document should be

signed or bear the signature of any person, then, notwithstanding anything contained in such law, such requirement shall be deemed to have been satisfied if such information or matter is authenticated by the means of digital signature affixed in such manner as may be prescribed by the Central Government.”

Electronic Record (Section 2(1) (t)). “Means data, record or data generated, image or sound stored, received or sent in an electronic form, or microfilm or computer generated micro-fiche.”

Legal recognition of Electronic Record (section 4). “Where any law provides that the information or any other matter shall be in writing or in typewritten or printed form, then, notwithstanding anything contained in such law, such requirement shall be deemed to have been satisfied if such information or matter is: (a) rendered or made available in an electronic form; and (b) accessible so as to be usable for a subsequent reference.”

Secure Electronic Record (Section 14). “Where any security procedure has been applied to an electronic record at a specific point of time, then such record shall be deemed to be a secure electronic record from such point of time to the time of verification.”

Secure Digital Signature (Section 15). “If, by application of a security procedure agreed to by the parties concerned, it can be verified that a digital signature, at the time it was affixed, was: (a) unique to the subscriber affixing it; (b) capable of identifying such subscriber; (c) created in a manner or using a means under the exclusive control of the subscriber and is linked to the electronic record to which it relates in such a manner that if the electronic record was altered the digital signature would be invalidated, then such digital signature shall be deemed to be a secure digital signature.”

Certifying Authority (Section (2(1)(g)). “Means a person who has been granted a licence to issue a Digital Signature Certificate under section 24” (issuance of certificates by Controller).

Treatment of Certification Authorities (Chapter VI). “This Act authorizes the Central Government to appoint a Controller of Certifying Authorities. The duties of the Controller are listed under Chapter VI of the Act, and include exercising supervision over the activities of certification authorities and defining the duties of these certification authorities.”

Towards wider adoption

The adoption of digital signatures in India is still at an early stage. Though the idea of digital signatures appears to be sound, it has not lived up to expectations. At the moment, applications of digital signatures are limited to sectors such as banking and financial services, online stock-

trading portals, and engineering conglomerates (to authenticate critical engineering drawings and documents).

Cyber Crime and Punishment :

The computer-generated world of internet is known as cyberspace and the laws prevailing this area are known as Cyber laws and all the users of this space come under the ambit of these laws as it carries a kind of worldwide jurisdiction. Cyber law can also be described as that branch of law that deals with legal issues related to use of inter-networked information technology. In short, cyber law is the law governing computers and the internet.

The growth of Electronic Commerce has propelled the need for vibrant and effective regulatory mechanisms which would further strengthen the legal infrastructure, so crucial to the success of Electronic Commerce. All these governing mechanisms and legal structures come within the domain of Cyber law.

Cyber law is important because it touches almost all aspects of transactions and activities and on involving the internet, World Wide Web and cyberspace. Every action and reaction in cyberspace has some legal and cyber legal angles.

Cyber Crime is not defined in Information Technology Act 2000 nor in the National Cyber Security Policy 2013 nor in any other regulation in India. Hence, to define cyber-crime, one can say, it is just a combination of crime and computer. In other words "any offence or crime in which a computer is used is a cyber-crime". Even a petty offence like stealing or pick pocket can be brought within the broader purview of cybercrime if the basic data or aid to such an offence is a computer or an information stored in a computer used (or misused) by the fraudster. The I.T. Act defines a computer, computer network, data, information and all other necessary ingredients that form part of a cybercrime.

SECTION PUNISHMENT

Section 43	This section of IT Act, 2000 states that any act of destroying, altering or stealing computer system/network or deleting data with malicious
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intentions without authorization from owner of the computer is liable for the payment to be made to owner as compensation for damages.

Section 43A	This section of IT Act, 2000 states that any corporate body dealing with sensitive information that fails to implement reasonable security practices causing loss of other person will also liable as convict for compensation to the affected party.
Section 66	Hacking of a Computer System with malicious intentions like fraud will be punished with 3 years imprisonment or the fine of Rs.5,00,000 or both.
Section 66 B, C, D	Fraud or dishonesty using or transmitting information or identity theft is punishable with 3 years imprisonment or Rs. 1,00,000 fine or both.
Section 66 E	This Section is for Violation of privacy by transmitting image or private area is punishable with 3 years imprisonment or 2,00,000 fine or both.
Section 66 F	This Section is on Cyber Terrorism affecting unity, integrity, security, sovereignty of India through digital medium is liable for life imprisonment.
Section 67	This section states publishing obscene information or pornography or transmission of obscene content in public is liable for imprisonment up to 5 years or fine or Rs. 10,00,000 or both.